Helmholtz's equation

Fredy W. Aquino

March 2, 2026

1 Statement of the problem

Solve Helmholtz's equation

$$\nabla^2\Phi + k^2\Phi = g \tag{1}$$

where Φ is the field quantity.

In particular, solve for the case $g = 0$ and unknown k: homogeneous, scalar Helmholtz's equation.

A nonuniform mesh is constructed as shown in Figure (1) where a rectangular solution region of size $a \times b$ is divided into rectangular elements, each of which is later divided into two triangular elements. Later, Φ will be evaluated using the computed mesh. The implementation only solve for TM modes (e.g. prescribed nodes $\Phi_p = 0$). In Figures (2) and (3) we show that the error from using FE to estimate k_c gets reduced upon increasing the number of elements n_x.

The analytical value for k_c is

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2} \tag{2}$$

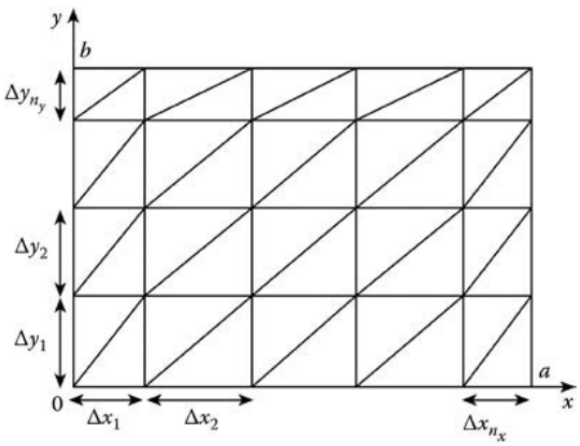

Figure 1: Discretization of a rectangular region into a nonuniform mesh

TABLE 6.5

Lowest Wavenumber for a Square Waveguide ($b = a$)

n_x	n_e	$k_c a$	% Error
2	8	5.656	27.3
3	18	5.030	13.2
5	50	4.657	4.82
7	98	4.553	2.47
10	200	4.497	1.22

Exact: $k_c a = 4.4429$, $n_y = n_x$.

Figure 2: Computing lowest (or dominant) cutoff wavenumber k_c of the TM_{11} mode in waveguides for square cross section ($a \times a$)

Lowest Wavenumber for a Rectangular Waveguide ($b = 2a$)

n_x	n_e	$k_c a$	% Error
2	16	4.092	16.5
4	64	3.659	4.17
6	144	3.578	1.87
8	256	3.549	1.04

Exact: $k_c a = 3.5124$, $n_y = 2n_x$.

Figure 3: Computing lowest (or dominant) cutoff wavenumber k_c of the TM_{11} mode in waveguides for rectangular cross section ($a \times b$)